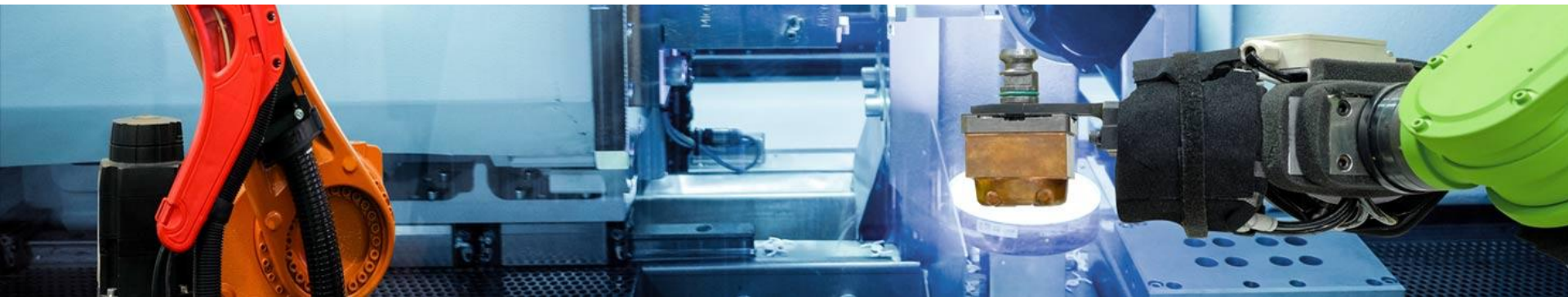
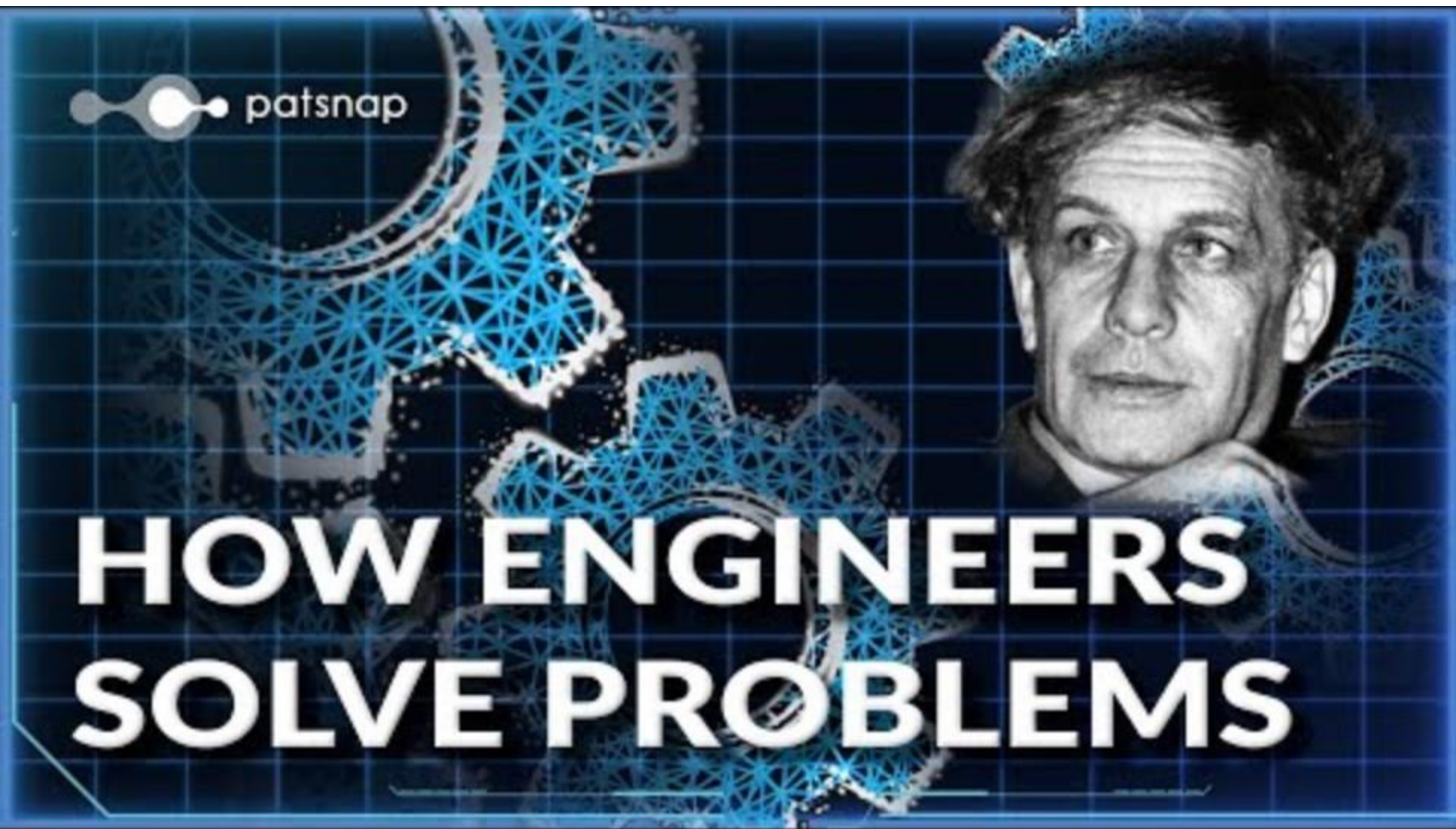




Mechatronic system design techniques

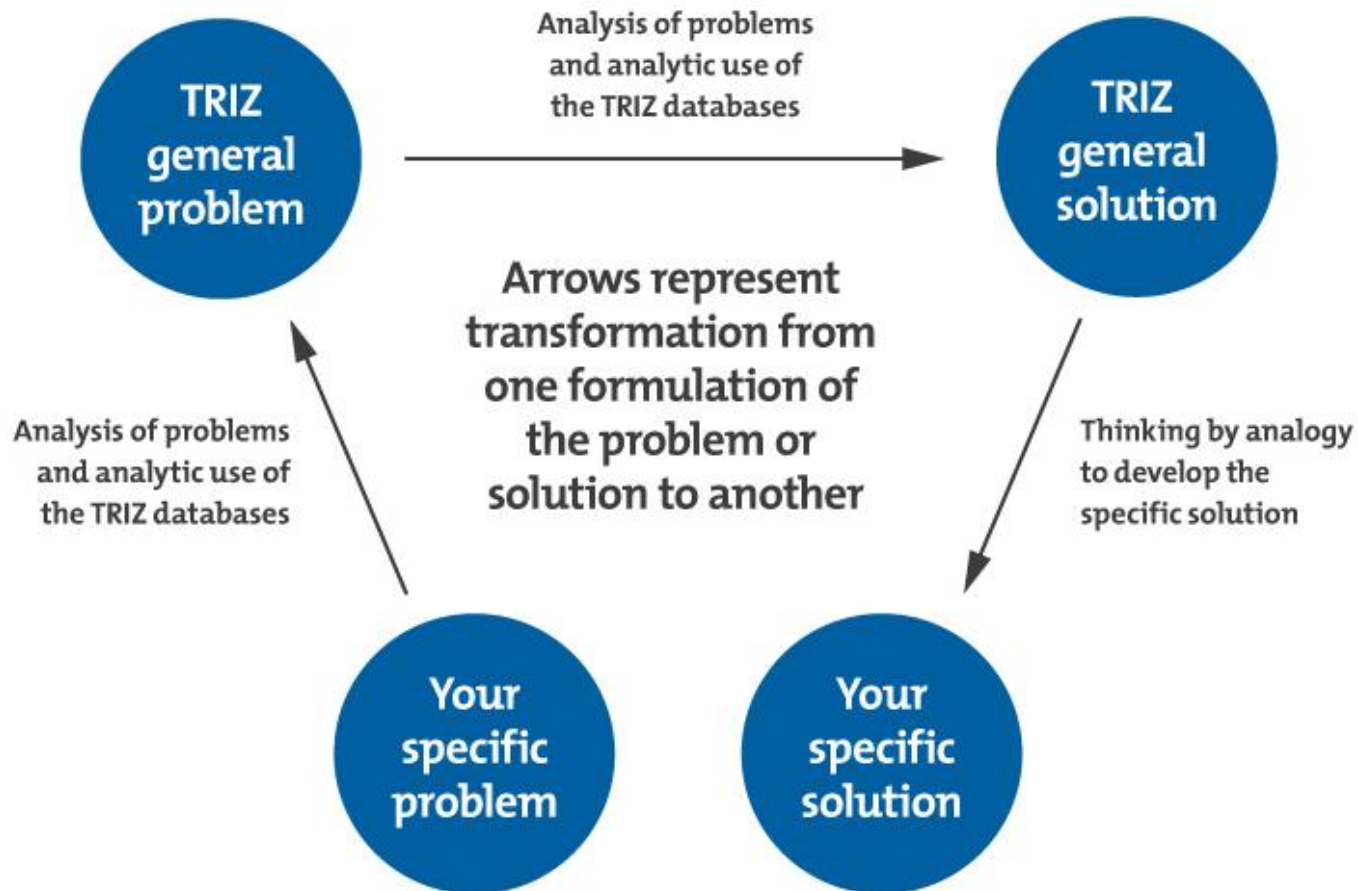


TRIZ



TRIZ

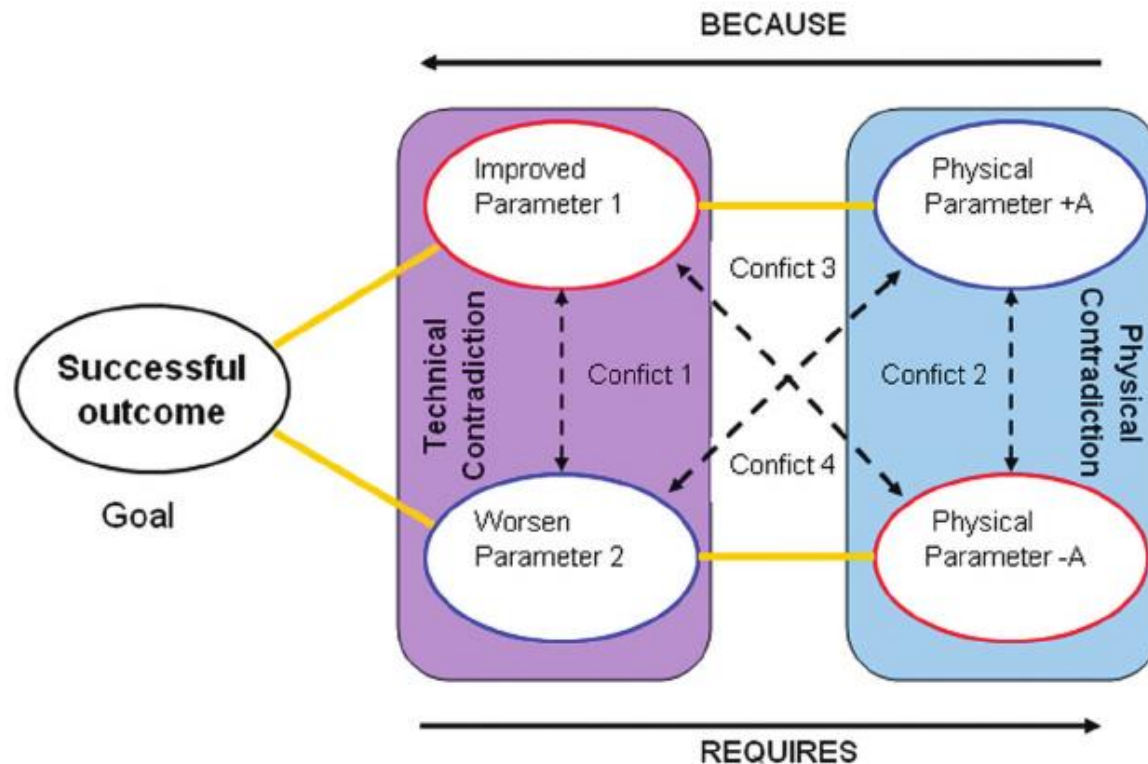
TRIZ (viết tắt của **Theory of Inventive Problem Solving** - Lý thuyết Giải quyết vấn đề sáng tạo) là một phương pháp phân tích và giải quyết vấn đề, được phát triển bởi nhà phát minh người Liên Xô Genrich Altshuller. TRIZ giúp các kỹ sư, nhà sáng tạo tìm ra các giải pháp đổi mới thông qua việc **nhận diện** và **giải quyết mâu thuẫn** trong các hệ thống kỹ thuật.



Các thành phần chính của TRIZ – Mâu thuẫn

Mâu thuẫn

- TRIZ nhấn mạnh rằng phần lớn các vấn đề kỹ thuật phát sinh từ các mâu thuẫn. Mâu thuẫn có thể chia làm hai loại chính:
 - **Mâu thuẫn kỹ thuật:** Khi cải thiện một đặc tính của hệ thống sẽ làm xấu đi một đặc tính khác.
 - **Mâu thuẫn vật lý:** Khi một phần tử của hệ thống cần phải có những thuộc tính đối lập (ví dụ: vừa nóng vừa lạnh cùng một lúc).
- TRIZ cung cấp các công cụ để giải quyết những mâu thuẫn này mà không phải hy sinh một yếu tố nào.



Các thành phần chính của TRIZ - 40 Inventive Principles

page A - 40 principles of invention in TRIZ method, sketches rendered into vector graphics

01) DIVISION a) a ship built, made of removable/replaceable bulkheads b) multi-engine aircraft c) multi-piston engine of internal combustion d) a toy made from Lego blocks e) breakable chocolate f) multi-grip gators g) a binded file of paper sheets h) multi-blade cardrags razors i) multi-blade airscrews of aircrafts, or wind power-plants j) fishing floats, floating beacons, etc.	08) ANTI-WEIGHT (balance preserving) a) wind turbines (moment of inertia in set of smoothing gusts of wind) b) anti-airscrews c) fish bladder (fish submerged in water) d) balloon filled with hot air e) slipping hydrofoils boats f) concept of hover crafts g) F-formation = $2MV^2/R$ h) $V = 2\pi R\omega$	15) DYNAMICS a) automatically extensible/opened doors, air-locks, etc., retracted when it is needed b) automatic gears in mobiles c) undercarriages in cars of variable stiffness characteristics, tuned exactly to terrain conditions during the driving d) electronic controllers for carburettor, electronically controlled fuel injection in dependency of driving conditions
02) TAKING OUT a) taking of notoriously noisy power unit, or compressor out of the main boat b) (engines, turbines, blades) combined with internal ducts for air ventilation system, taken out of the building, i.e. placed on the buildings elevations c) sound of bird's predator, previously registered on a tape, and played back, can be used scaring away the birds, notoriously flying near or around the airports	09) PRE-ELIMINARY ANTI-ACTION (COUNTER-ACTION) a) surrounding sounds b) counter-acting active earphones c) piezoelectric anti-impact system for cutting tool	16) EXCESSIVE (OR PARTIAL) ACTION a) in close fit of both piston and cylinder of the engine b) material allowance left in fit on inner side of cylinder designed for lapping c) to spray excessively paint, and then to remove the excess of the paint d) to fulfill the fuel tank, and then to remove the excess of fuel
03) LOCAL QUALITY a) dustless excavation of coal - the dust is captured by tiny droplets inside of the water cone b) bigger droplets outside of the cone keep the dust in a place c) weighed estimation produced for rankings of computers, printers, etc.	10) PRE-ELIMINARY ACTION a) parking of hard disc reading/writing heads (when it is needed) b) blowing off of the (potentially choked) nozzles in printing cartridges c) (operational mode)	17) ANOTHER DIMENSION a) two colliding points in 1D should be rearranged in 2D plane b) two colliding points on 2D plane should be rearranged into 3D space c) use of Lie algebra, in analysis of structures of atoms d) in studying of electron shells, around atomic nuclei e) science of complex asymmetries in crystallography
04) ASYMMETRY a) pneumatic tyre asymmetrically reinforced from outside, due to contact with pavement curb b) left- or right-handed rules of priority, in right of road c) slanted concrete mixer, blender d) asymmetrically built conjunctions, handles e) asymmetrically defined functionality of the "trap-the-door" mechanisms f) asymmetrically built car, due to either left- or right-sided driver's sit	11) BEFOREHAND CUSHIONING a) for instance: a method of "dressing" of the cut tree branches (this action, actually forces a tree to beforehand react, to gather healing substances), a pressure band b) driver's airbag c) masking of the chosen elements, within patches on the object, before its painting	18) MECHANICAL SELF-INDUCED VIBRATIONS (IN RESONANCE) a) piezoelectric engine - a conceptual design b) electric circuit of inducting c) spring based tighteners for set of two discs d) quartz generators, in electric circuits e) both sided metal plating of layer ceramic material fasten to lower disc
05) MERGING a) several computers combined into functioning network b) a hedge made of pales c) textiles made of wool/polyester/cotton fibres d) roofing tiles combined into coverage of house roof e) mobile concrete mixer, mobile crane, refrigerator, merged into single mobile machine unit, combining of the stationary machines with mobile undercarriages	12) EQUIPOTENTIALITY a) a sequence of linear movements is replaced by single seamless movement on section of arc b) a heavy element of the press, lifted up, and carried away usually in sequence of linear movements, is replaced with press deflected on remotely fasten long arm c) a press deflected from position d) a press deflected on long arm e) dissolvable surgeon threads f) rather to cool down stuck inner object, than to heat up other bigger outer object, which seizes the former one	19) PERIODICAL ACTION, OR PULSED ACTION a) hammer, drill b) pulse amplifiers c) pulsed laser, against layers of continuous operational mode d) "pseudo-analogue" driving (PWM) mapping (Width Modulation) e) pulse DC power unit, against conventional DC power unit f) step motors
06) UNIVERSALITY a) a helmet in use, within field conditions, rendered as a "handy-tools" b) universal "handy-tools" c) a) spade d) a2) frying pane e) universal kitchen robots, mixers, blenders, with operating actuators (rasps, juice extractors, etc.)	13) INVERSION (UPSIDE DOWN) a) for instance: in reversing the working mode of vacuum cleaner, then, vapour could be used in cleaning of carpets b) to turn mounted object upside down, on assembling line c) turning (object in move, while motionless turning tool), against milling (mobile milling cutter) d) binary tree's structure is sought from root to leaves in (one in-depth) search algorithms, while another algorithm seeks through nodes from leaves to root	20) CONTINUITY ACTION OF USEFUL ACTION a) enlarging drill, operating in both directions b) nozzles of cartridge, printing also in returning direction (without idle mode) c) steam turbines of generators for one power plants, working continuously in optimal mode, while the others working, as pump-storage power plants, in aim of storing of energy for afternoon hours d) water in upper reservoir on mornings, while emptying upper reservoir into lower one on afternoons
07) EMBEDDED STRUCTURES (nested "Dolls" - Matryoshka) a) radiators of radio antennae b) refractive laser ring etc. c) radiators of ultrasound welders	14) SPHEROIDALITY, CURVATURES a) applications of b) replacement of linear movements by circular movements c) bearing rollers, spirals, shafts d) spheres, semi-domes e) application of arcs in architecture f) semi-domes in vaults of building g) circular accelerators (synchrotrons/magnetrans) h) in place of concept of linear separated fields of particles i) extensible, retractable measuring tape	21) SKIPPING, QUICK MODE, OR PACE OF REALIZATION a) wood-boring materials in quick thermal processing b) laser treatments of biological tissues c) in processing of hardy processed materials (both extremely soft and extremely hard) without thermal deformations, scorching, burnings d) picosecond pulsed lasers (femtosecond lasers) against laser of micro- and nano-seconds pulses e) various materials virtually have been separated, while treated with picosecond pulsed laser beam of energy f) steel hardening process in abrupt temperatures changes

A technique of Engineering Creativity and Inventive Thinking in Problem Solving

page B - 40 principles of invention in TRIZ method, sketches rendered into vector graphics

22) "BLESSING IN DISGUISE" (CONVERT HARM INTO BENEFIT) a) burning out main fire in outskirts b) to blow out the outskirts fire from the top of the main fire, to cut off fire's fuel c) permafrost materials are to be "treated" with liquid nitrogen d) the material's permafrost rapidly "liquefies"	28B) SUBSTITUTING OF MECH. SYS. WITH ELECTRO-MAGN. SYSTEMS A) magnetic borne pressure of the machined materials B) pneumatic magnetic pressure cushion C) mobile fields instead of static fields D) heterogeneous fields	34) DISCARDING & RECOVERING, (REJECT & PARTS REGENERATION) a) dissolvable medication capsules made of (biologically inert material) b) rocket's stages subsequently discarded during the flight c) cornstarch-based packages for dry products
23) FEEDBACK PRINCIPLE a) basically, as well as particularly: input signal → object → output signal (indoor or temperature regulation) closed loop with negative feedback (temperature) b) autopilot provided with 3-axis gyro system c) robot arms movements back-controlled in set of: 1) diode - 2) photodiode - 3) semi-transparent either: protractor, or: linear scale - placed in between	29A) PNEUMATICS & HYDRAULICS - pneumatic automobile tyre, - pneumatics (air-tight) dampers, automobile airbags, - pneumatic "discrete", driving of operational actuators, for instance: in automatic welding of packaging covers made of plastic wrapping on the figure above, in blue: approximate section of automobile pneumatic tyre	35) CHANGING STATE, PARAMETERS, PROPERTIES OF MATERIALS 1) high temperature food processing 2) low-temperature food preserving 3) a product ready for further processing step (for submerging in liquid chocolate) temp > 0°C temp < 0°C
24) INTERMEDIATE MEANS, "FITTING" PRINCIPLE a) in electronic circuits fitting either of: - impedance, - or resistance, - or input source to receiver b) fitting in mean of: - pressure-flowing (fluid mechanics), - loading of force moments, in transition gears (mechanical fitting) - stress of two interfacing surfaces (endurance)	29B) PNEUMATICS & HYDRAULICS - automobile brakes, - in driving of plane elevator, where the precision of driving is needed, as well as enormous force transition c) hydraulics in communicating vessels d) brake's block $F_1 = F_2$ $S_1 = S_2$ $F_2 > F_1$	36) PHASE TRANSITION a binary, phase transition cycle for refrigerator construction b) heat flows from surroundings c) heat flows directed to blue heat exchanger d) liquefied ammonia, heat carrier, (freon, ammonia, etc.) e) circulation of an external fluid in heat exchanger temp > 0°C temp < 0°C
25) SELF-SERVICING PRINCIPLE a) self-servicing system b) constant regeneration of the glow of halogen lamp c) tungsten sublimates to halogens then, redepot on tungsten glow	30) FLEXIBLE FILMS, FOILS, MEMBRANES a) not wettable film prohibits evaporation of water b) wrapping packaging based on plastic air-pumped bubbles c) flattable balloons, domes, barriers	37) THERMAL EXPANSION 1) thermal shift fitting 2) state of thermal balance temp > 0°C temp < 0°C
26) COPYING, IMAGING PRINCIPLE (application of optical mapping) a) use of ultrasonics b) magnetic resonance c) X-rays radiography d) in mapping of material structures the application of: - infrared - ultraviolet - basically of optical methods e) use of fluorescence and of scintillation's materials	31) POROUS MATERIALS a) aerated concrete (porous concrete) b) polyurethane foam c) catalysing surfaces in chemistry d) "vacuum" as a "construction" (building) process, draining of absorbing heat, collecting and depositing of chips e) porous abrasive tools f) porous sponge materials in kitchen getters g) porosity of grinding wheel h) porosity of grinding wheel i) porosity of grinding wheel j) porosity of grinding wheel	38) STRONG OXIDANTS a) oxygen b) ozone c) (indirectly) vapour H ₂ O d) in oxidation of metal's surface (iron with over-heated vapour under pressure, at 300°C degree) e) the surface with protection layer obtained due to oxidation
27) INEXPENSIVE SHORT-LIVED OBJECTS (CHEAP CADUCITY, & OF DISPOSABLE MATERIALS) a) kitchen utensils, disposable syringes, cutlery made of plastic b) disposable gloves, etc. c) plastic bags, packaging wrappers, etc. d) printing head integrated with ink cartridge (formerly, each printer possessed built-in printing head, presently, each of ink cartridge has its own printing head)	32) COLOUR CHANGING (ALTERNATING) a) in lapping process for inner surfaces of engine pistons & cylinders, the probing of phosphorescence distribution can be used	39) NEUTRAL ATMOSPHERES, INERT ENVIRONMENTS a) C ₆₀ extinguishers b) N ₂ or He ₂ protection atmospheres in processing, and production c) N ₂ or He ₂ protection atmospheres in storing of products, and materials, both raw and processed
28A) PRINCIPLE OF SUBSTITUTING OF MECHANICAL SYSTEM WITH FUNCTIONALLY EQUIVALENT ELECTRO-MAGNETIC SYSTEMS a) electric field b) magnetic field c) mechanical pressure or fastening	33) HOMOGENEITY the two interfacing surfaces should be made of the same material moreover, the similarities can be applied, regarding: - comparable mat's hardness, chemical inertness, structures - comparable thermal expansion's coefficients, (in case of dental materials; metal-glass conjunctions), - comparable electro-chemical potentials (in avoiding electro-chemical bone corrosion) - same fatigue characteristics, and amortization specifics	40) COMPOSITE MATERIALS 1) elements of blades, rotors, airnozzles in wind turbines constructions; 2) yacht's & catamaran's constructions; 3) elements exposed to ultra-strong, severe stress

A technique of Engineering Creativity and Inventive Thinking in Problem Solving

Các thành phần chính của TRIZ Contradiction Matrix



www.systematic-innovation.com

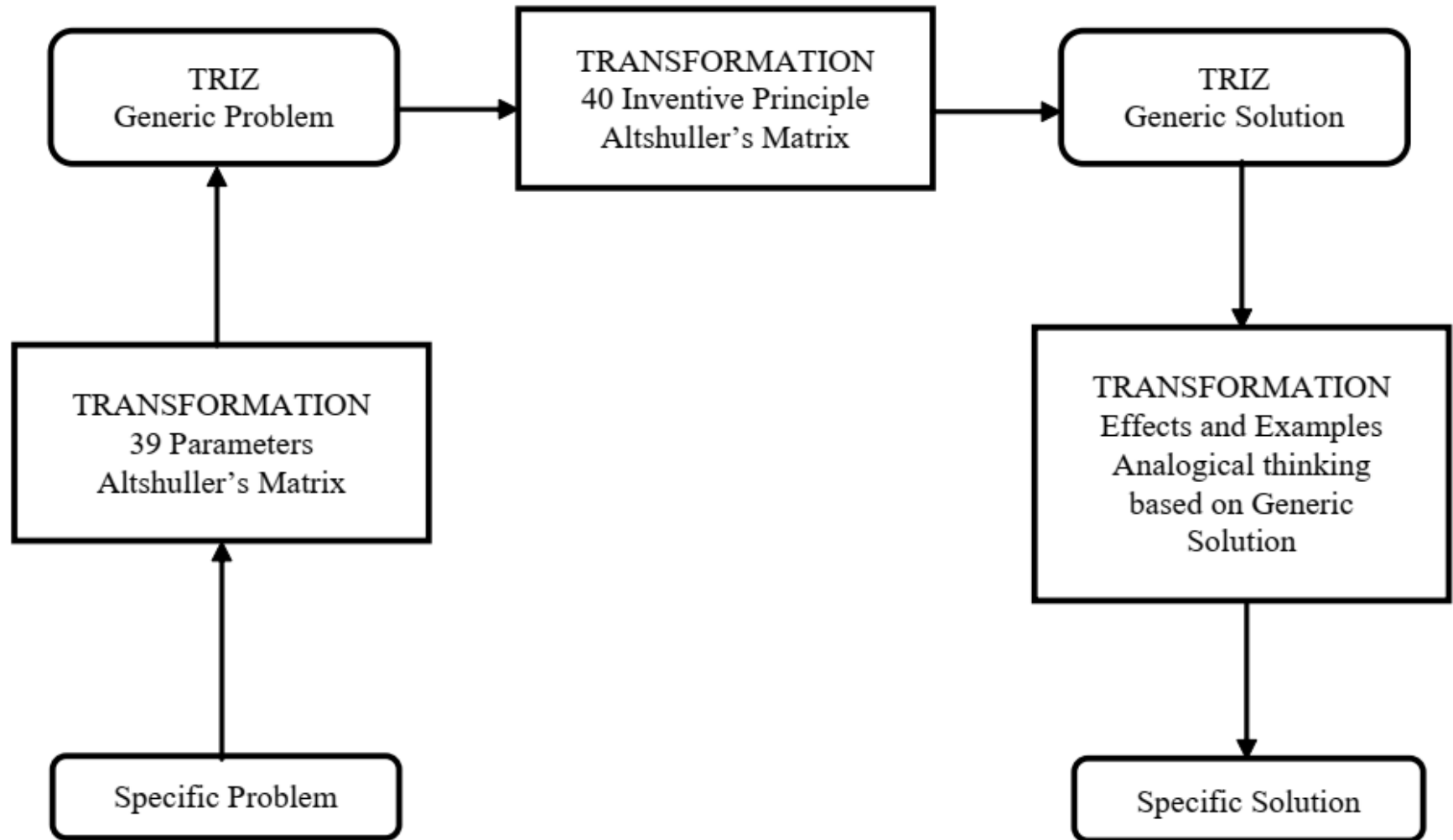
40 Inventive Principles

- 1 Segmentation
- 2 Taking Out/Separation
- 3 Local Quality
- 4 Asymmetry
- 5 Merging
- 6 Universality
- 7 'Nested Doll'
- 8 Anti-Weight
- 9 Preliminary Anti-Action
- 10 Preliminary Action
- 11 Beforehand Cushioning
- 12 Equipotentiality
- 13 'The Other Way Around'
- 14 Curvature
- 15 Dynamization
- 16 Partial Or Excessive Actions
- 17 Another Dimension
- 18 Mechanical Vibration
- 19 Periodic Action
- 20 Continuity Of Useful Action
- 21 Skipping
- 22 'Blessing In Disguise'
- 23 Feedback
- 24 Intermediary
- 25 Self-Service
- 26 Copying
- 27 Cheap Short-Living Objects
- 28 Mechanics Substitution
- 29 Pneumatics And Hydraulics
- 30 Flexible Shells And Thin Films
- 31 Porous Materials
- 32 Colour Changes
- 33 Homogeneity
- 34 Discarding And Recovering
- 35 Parameter Changes
- 36 Phase Transitions
- 37 Thermal Expansion
- 38 Strong Oxidants
- 39 Inert Atmosphere
- 40 Composite Materials

Feature	Worsening Feature	Physical										Performance										Efficiency										ility										Manufacture/Cost										Measure-ment																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
		Weight of Moving Object	Weight of Stationary Object	Length/Amount of Moving Object	Length/Amount of Stationary Object	Area of Moving Object	Area of Stationary Object	Volume of Moving Object	Volume of Stationary Object	Shape	Amount of Substance	Amount of Information	Amount of Energy	Amount of Power	Speed	Force/Torque	Energy Used by Moving Object	Energy Used by Stationary Object	Power	Stress/Pressure	Strength	Stability	Illumination Intensity	Function Efficiency	Loss of Substance	Loss of Time	Loss of Energy	Loss of Information	Noise	Harmful Emissions	Other Harmful Effects Generated by System	Adaptability/Variability	Compatibility/Connectability	Transferrability/Operability	Reliability/Robustness	Reparability	Security	Battery/Vulnerability	Aesthetics/Appearence	Other Harmful Effects Generated by System	Manufacturability	Manufacture Precision/Consistency	Automation	Productivity	System Complexity	Control Complexity	Ability to Detect/Measure	Measurement Precision																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
1	Weight of Moving Object	3	19	17	15	17	28	28	40	35	3	35	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
2	Weight of Stationary Object	30	3	2	35	9	17	15	17	31	14	14	13	31	35	13	35	40	35	35	3	35	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13	14	10	13

TRIZ Method

https://www.triz40.com/aff_Principles_TRIZ.php



TRIZ Method – Case study

Resolving the passive compliant gripper design contradictions by TRIZ

1. Specific Problem -> TRIZ Generic Problem:

- Vấn đề cụ thể là thiết kế một gripper robot thích ứng thụ động.
- Tác giả đã chuyển đổi vấn đề này thành các vấn đề tổng quát TRIZ như tính thích ứng, độ phức tạp của thiết bị, và độ tin cậy.

2. TRIZ Generic Problem -> TRIZ Generic Solution:

- Sử dụng ma trận mâu thuẫn của Altshuller với 39 tham số kỹ thuật.
- Xác định được 4 nguyên tắc sáng tạo chính từ 40 nguyên tắc của TRIZ: Dynamics, Local quality, Parameter changes, và Universality.

Feature to improve	Undesired result (degraded feature)		
	33. Ease of operation	36. Complexity of device	37. Complexity of control
35. Adaptability of versatility		15, 28, 29, 37	
38. Level of automation	1, 3, 12, 34		
27. Reliability		1, 13, 35	
32. Ease of manufacture			1, 6, 11, 28

TRIZ Method – Case study

Resolving the passive compliant gripper design contradictions by TRIZ

TRIZ Generic Problem

- + **Mâu thuẫn kỹ thuật 1:** Nếu cải thiện khả năng thích ứng, điều này có thể làm cho thiết bị trở nên phức tạp hơn.
 - + **Mâu thuẫn kỹ thuật 2:** Khi tăng mức độ tự động hóa, có khả năng làm giảm sự dễ dàng trong vận hành của thiết bị.
 - + **Mâu thuẫn kỹ thuật 3:** Việc tăng độ tin cậy có thể làm tăng độ phức tạp trong điều khiển hoặc làm tăng chi phí sản xuất.
- => Bằng cách sử dụng Ma trận TRIZ từ bước 2, nhóm nghiên cứu đã chuyển vấn đề cụ thể thành **mâu thuẫn kỹ thuật tổng quát**:

- Làm thế nào để **tăng cường khả năng thích ứng** mà không làm **tăng độ phức tạp**?
- Làm thế nào để **tăng mức độ tự động hóa** mà vẫn đảm bảo **dễ dàng vận hành**?
- Làm thế nào để **tăng độ tin cậy** mà không làm **tăng độ phức tạp của điều khiển** hoặc **chi phí sản xuất**?

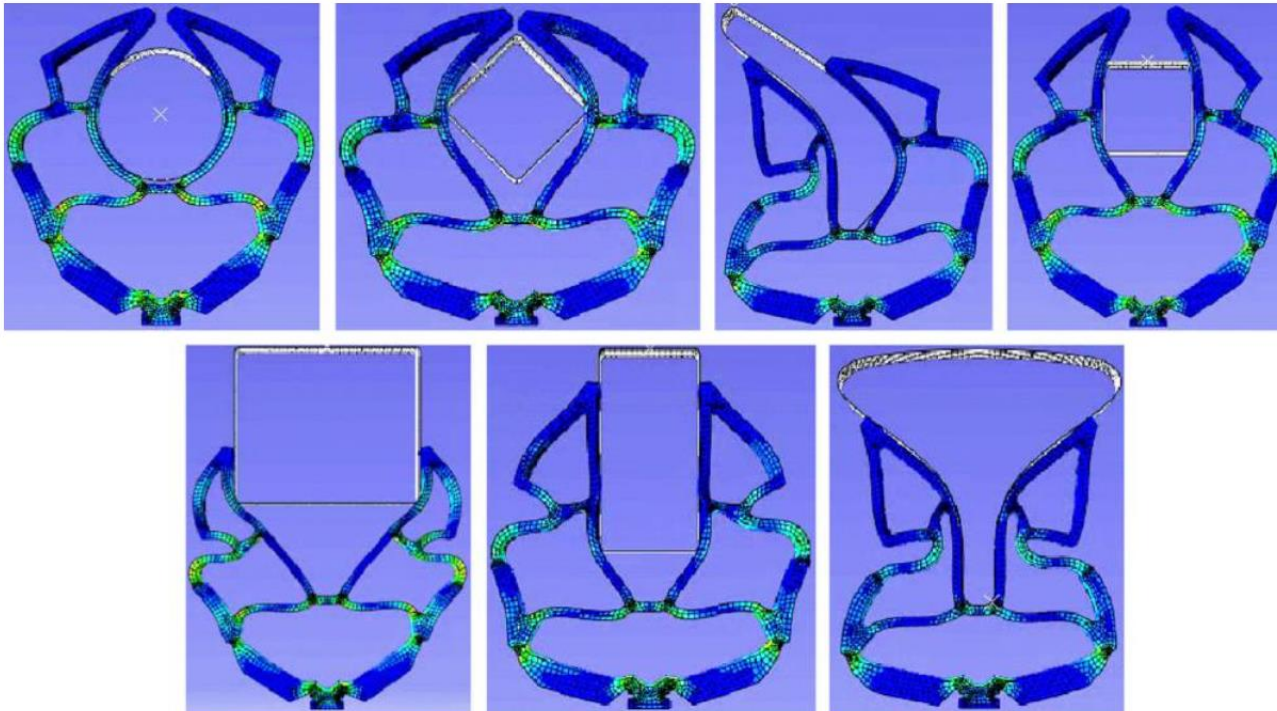
4. TRIZ Generic Solution -> Specific Solution:

- Áp dụng 4 nguyên tắc sáng tạo đã xác định để đưa ra giải pháp cụ thể:
 - Dynamics: Thiết kế gripper có cấu trúc đàn hồi phân bố.
 - Local quality: Tích hợp cảm biến vào cấu trúc gripper.
 - Parameter changes: Thử nghiệm các loại cao su silicone khác nhau.
 - Universality: Các bộ phận (cảm biến) thực hiện nhiều chức năng.

TRIZ Method – Case study

Ý nghĩa của nguyên tắc "Dynamics":

- Cho phép (hoặc thiết kế) các đặc tính của đối tượng, môi trường bên ngoài, hoặc quá trình thay đổi để đạt được trạng thái tối ưu hoặc tìm điều kiện hoạt động tối ưu.
- Chia đối tượng thành các phần có khả năng chuyển động tương đối với nhau.
- Nếu một đối tượng (hoặc quá trình) cứng hoặc không linh hoạt, hãy làm cho nó có thể di chuyển hoặc thích ứng được.

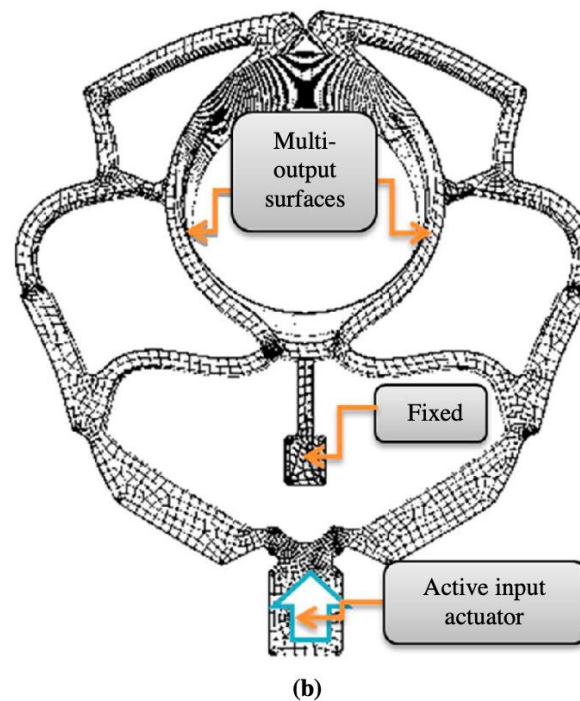
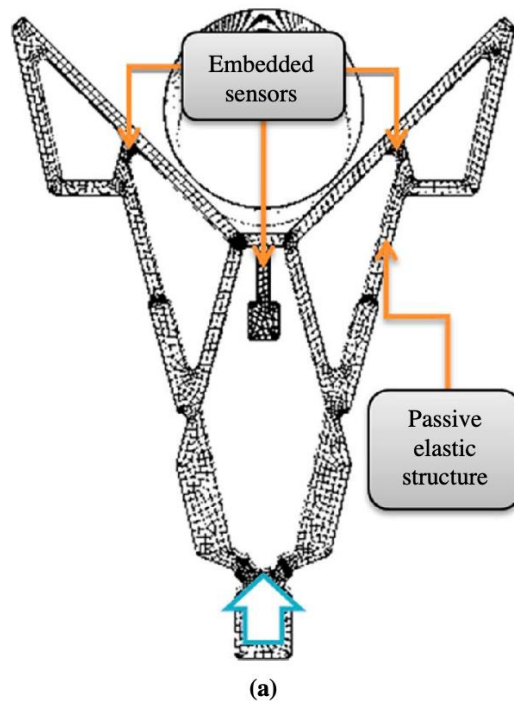


Verification of the gripper functions for different shapes of grasping objects

TRIZ Method – Case study

Ý nghĩa của nguyên tắc "Local quality":

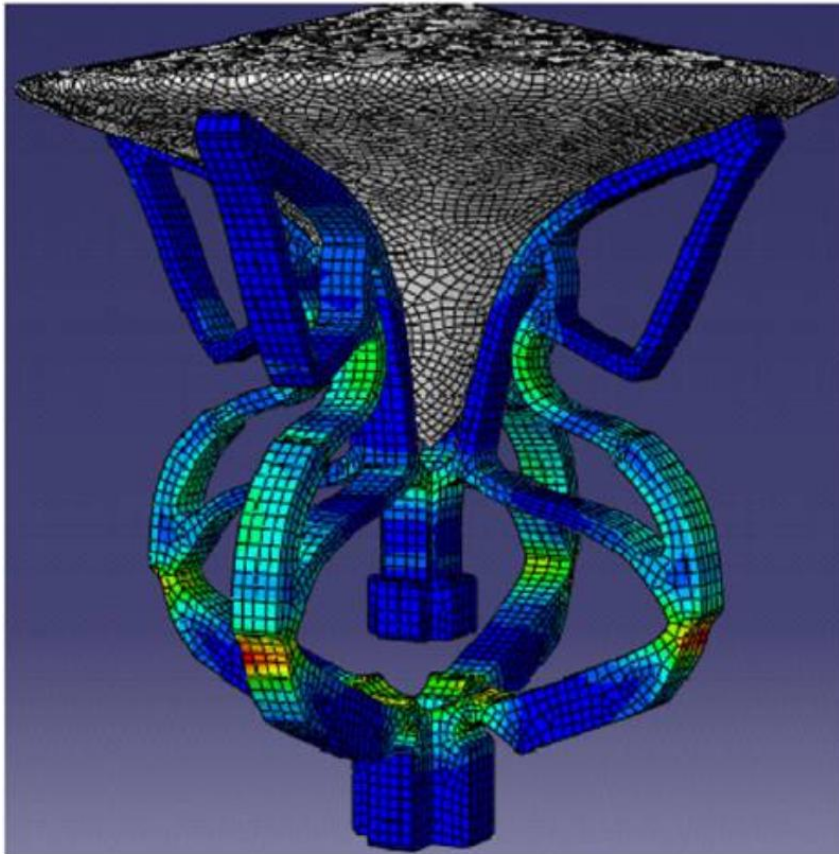
- Thay đổi cấu trúc của một đối tượng từ đồng nhất sang không đồng nhất.
- Thay đổi môi trường bên ngoài (hoặc ảnh hưởng bên ngoài) từ đồng nhất sang không đồng nhất.
- Làm cho mỗi phần của đối tượng hoạt động trong điều kiện phù hợp nhất với chức năng của nó.
- Làm cho mỗi phần của đối tượng thực hiện một chức năng khác nhau và hữu ích.



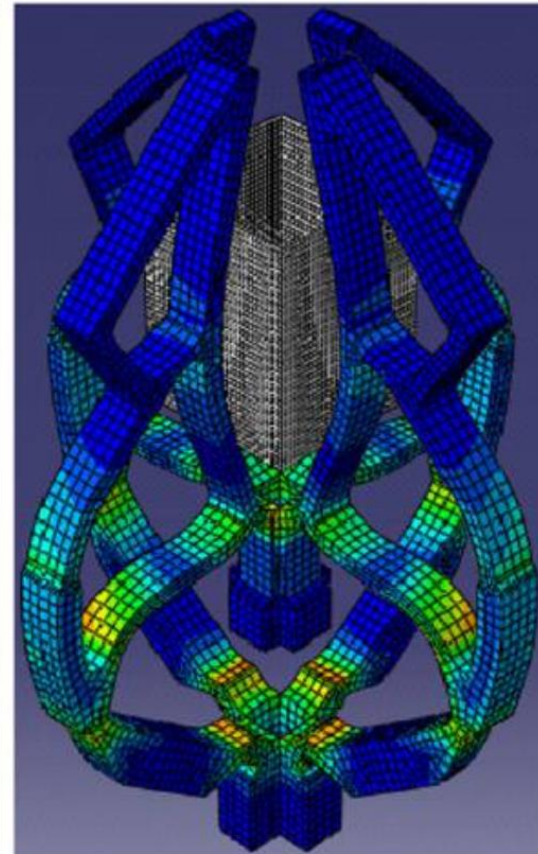
TRIZ Method – Case study

Ý nghĩa của nguyên tắc "Parameter changes":

- Thay đổi trạng thái vật lý của một đối tượng (ví dụ: từ khí sang lỏng hoặc rắn).
- Thay đổi nồng độ hoặc độ đặc.
- Thay đổi mức độ linh hoạt.
- Thay đổi nhiệt độ.



(c)

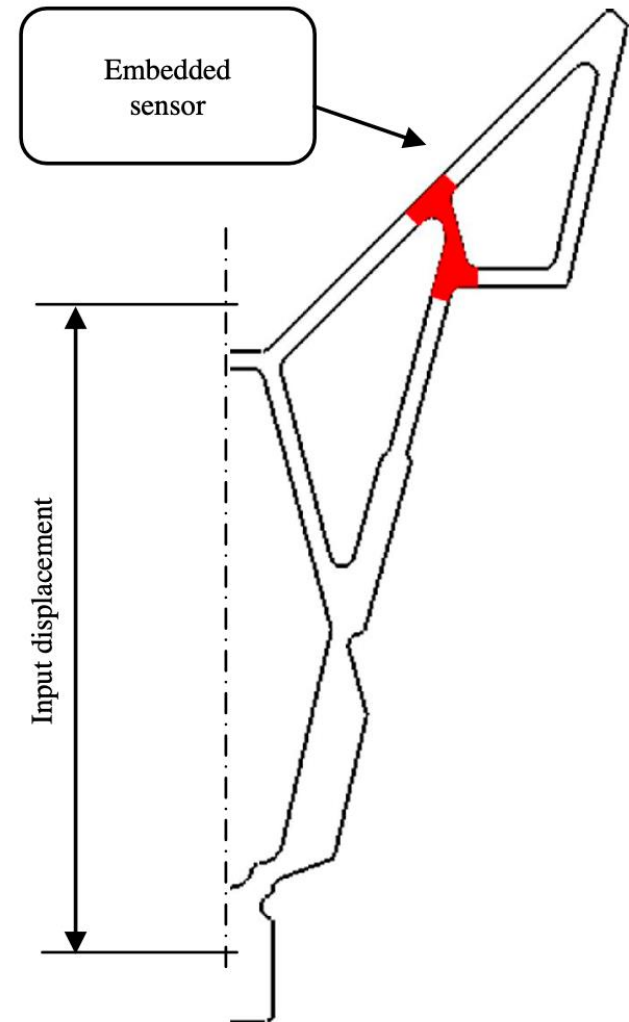
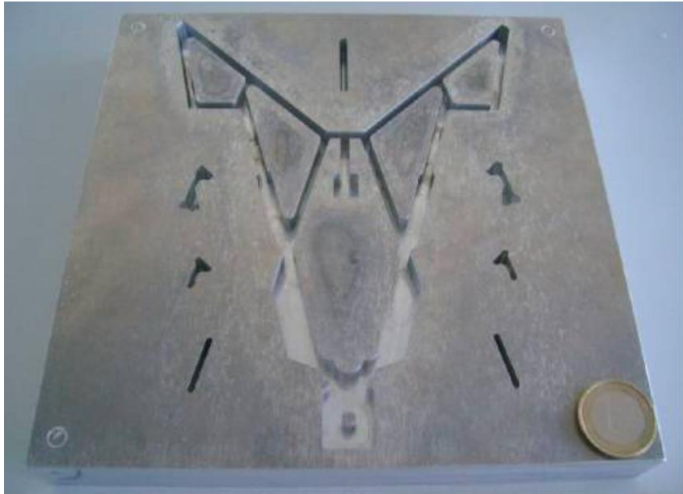


(d)

TRIZ Method – Case study

Ý nghĩa của nguyên tắc "Universality":

- Làm cho một phần hoặc đối tượng thực hiện nhiều chức năng.
- Loại bỏ nhu cầu sử dụng các phần khác.



Case study: Conceptual design for an In-Pipe Robot

Vấn đề cụ thể: Thiết kế một robot di chuyển trong ống với ít **động cơ** hơn và **điều khiển đơn giản hơn**

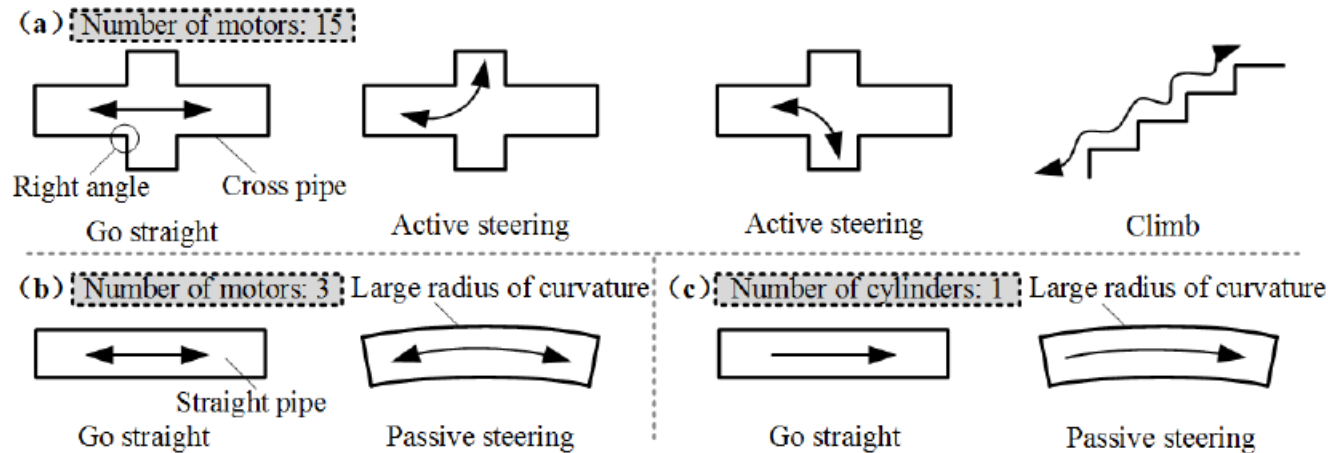


Figure 1. Move direction classification of peristaltic robots: (a) multi-direction, (b) two-direction, (c) one-direction.

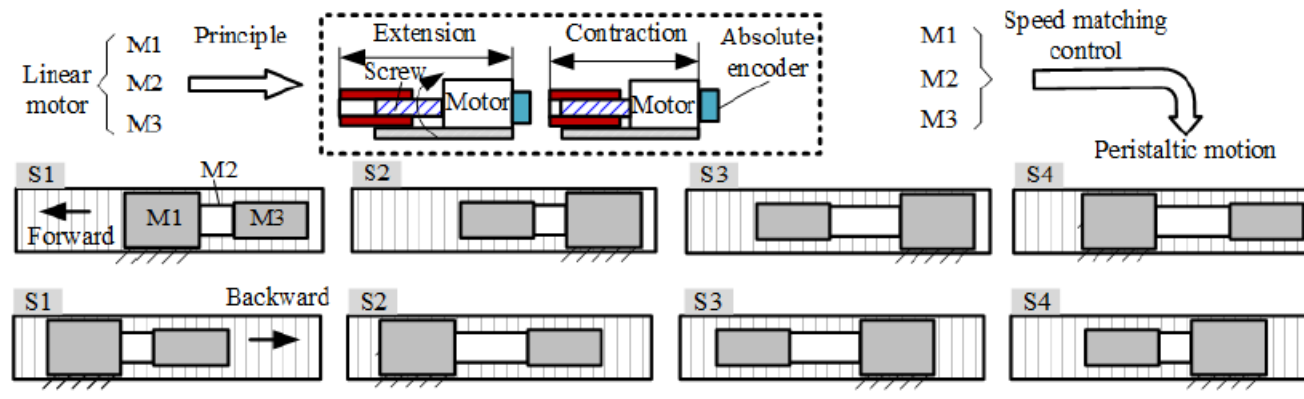


Figure 2. Traditional two-direction peristaltic robot driven by 3-motor structure (move forward and move backward).

Case study: Conceptual design for an In-Pipe Robot

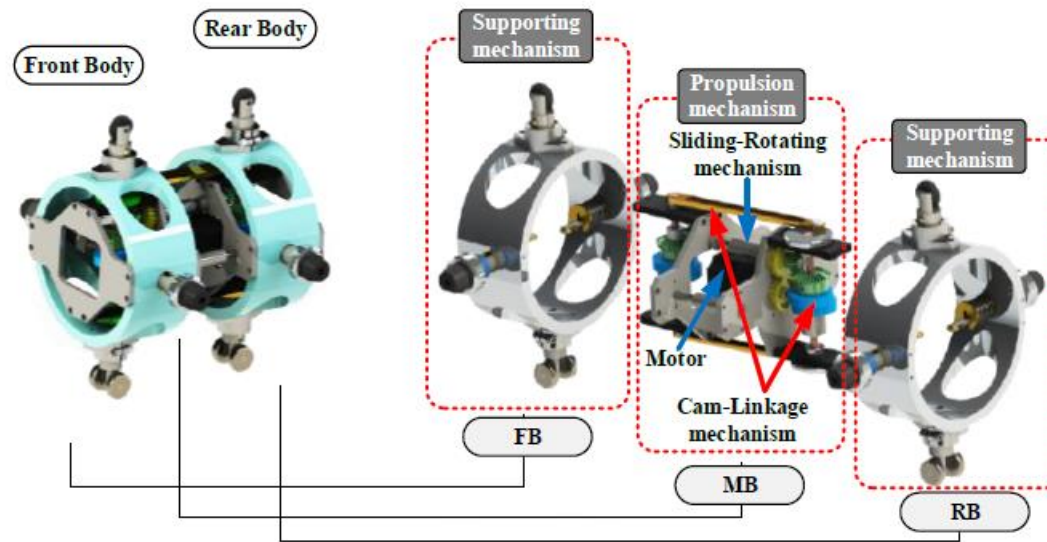


Figure 3. 3D rendering of the novel peristaltic in-pipe robot.

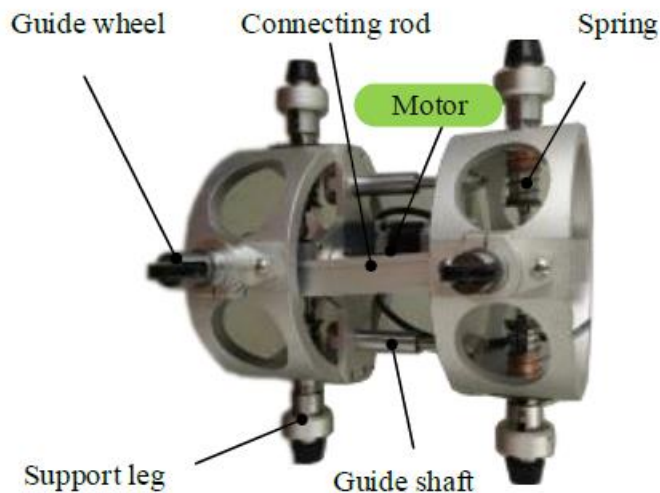


Figure 4. Novel in-pipe robot prototype.

Case study: Conceptual design for an In-Pipe Robot

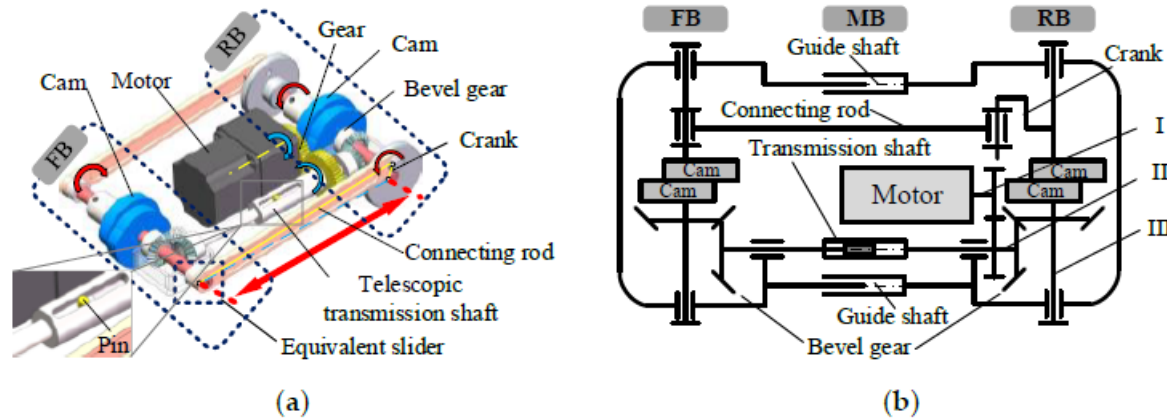


Figure 5. Propulsion mechanism of the novel in-pipe robot. (a) 3D; (b) 2D.

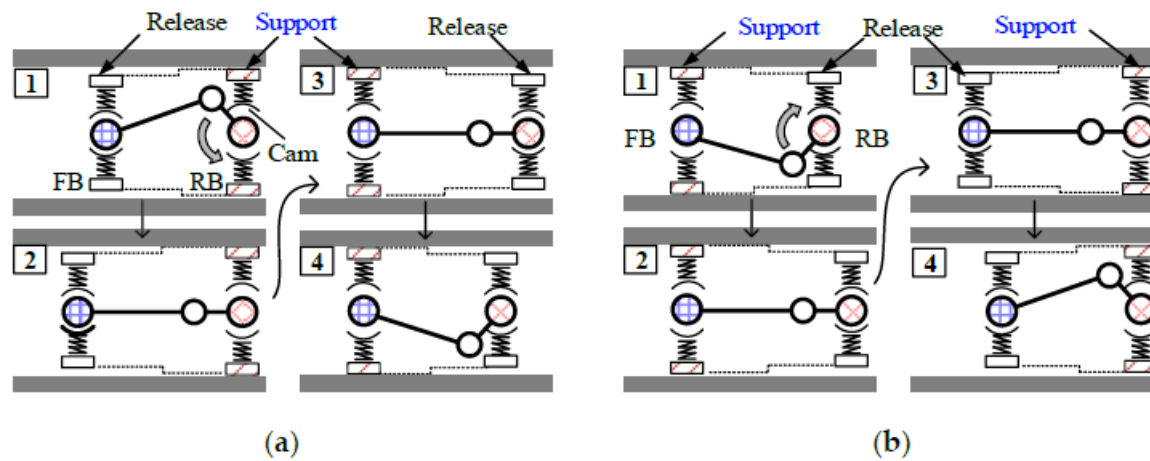


Figure 6. Principle of forward and reverse walking: (a) forward walking; (b) reverse walking.

Case study: Conceptual design for an In-Pipe Robot

- **Finding Design Conflicts**

A specific problem can be seen in terms of conflicts, and these conflicts can be further divided into physical conflicts and technical conflicts.

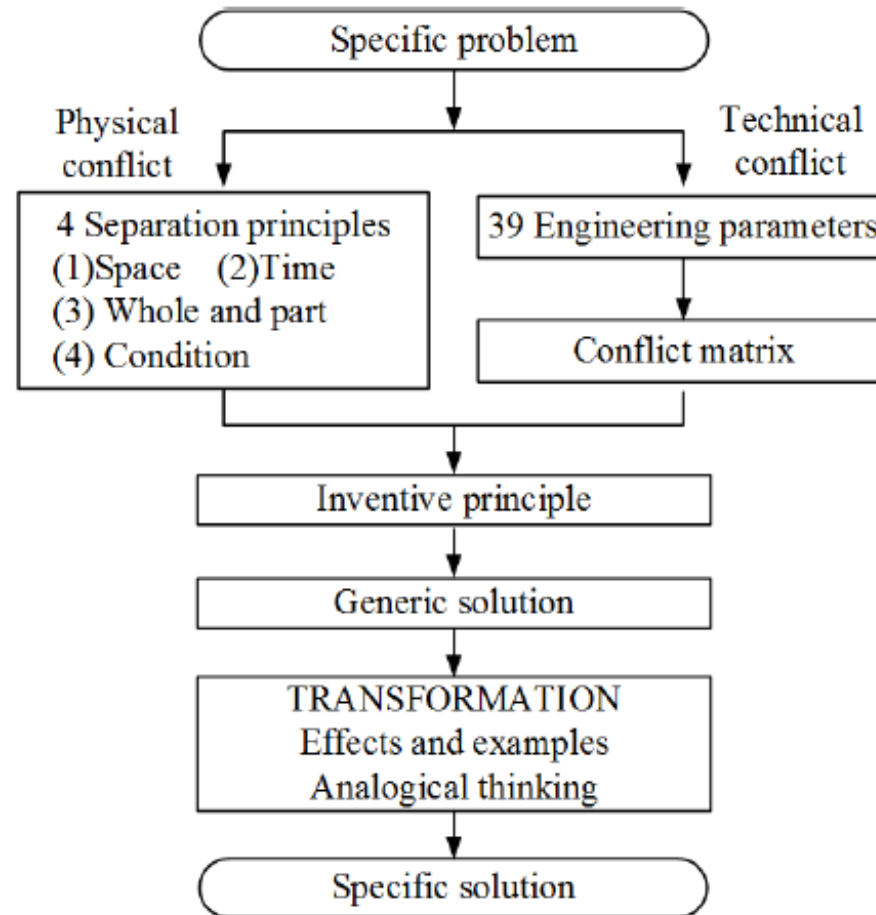
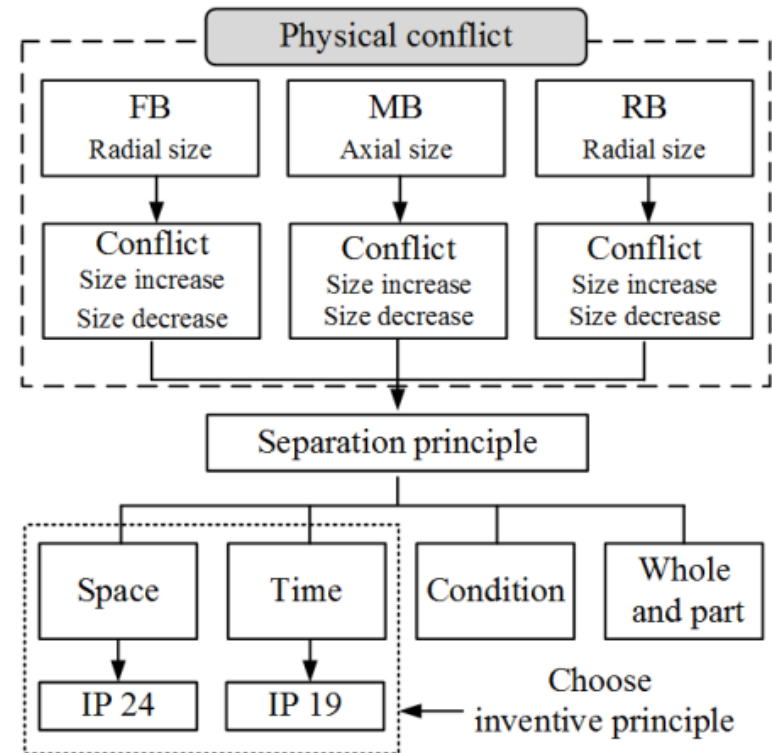
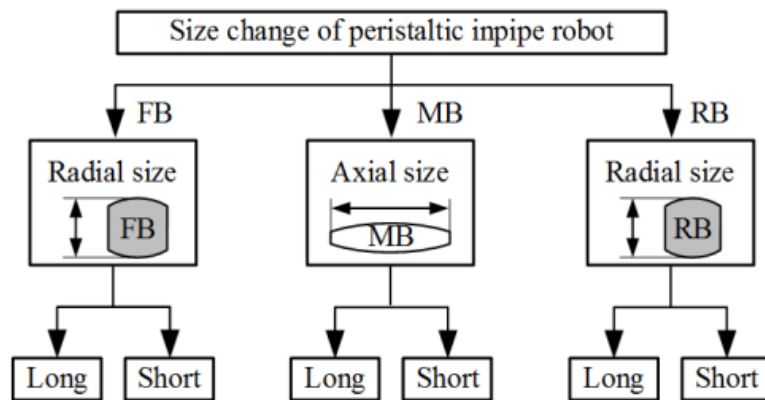


Figure 7. Conflict classification and resolution process.

Case study: Conceptual design for an In-Pipe Robot

Physic conflicts



Technical conflicts

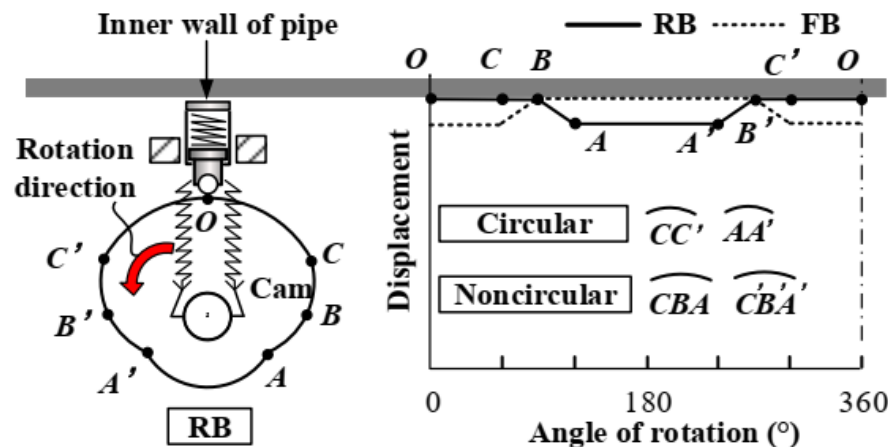
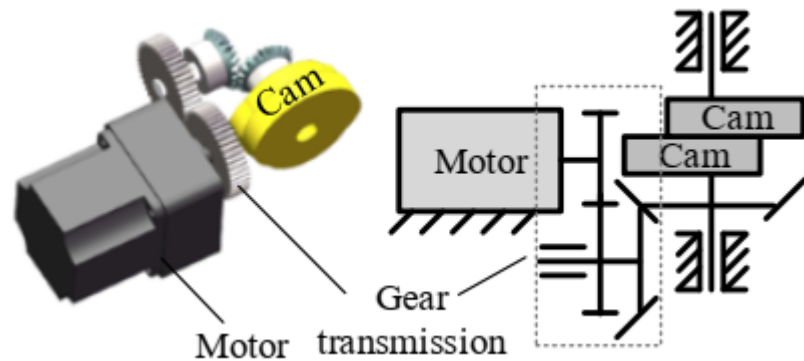
Improved Parameters	Deteriorating Parameter	Inventive Principle
37	35	1, 15
	32	5, 28, 11, 29

Case study: Conceptual design for an In-Pipe Robot

❑ Resolving Physic conflicts

Nguyên tắc sáng tạo 19 (Tác động chu kỳ): Thay thế chuyển động liên tục bằng chuyển động chu kỳ.

Giải pháp: Sử dụng cơ cấu cam để tạo ra chuyển động chu kỳ cho FB và RB.



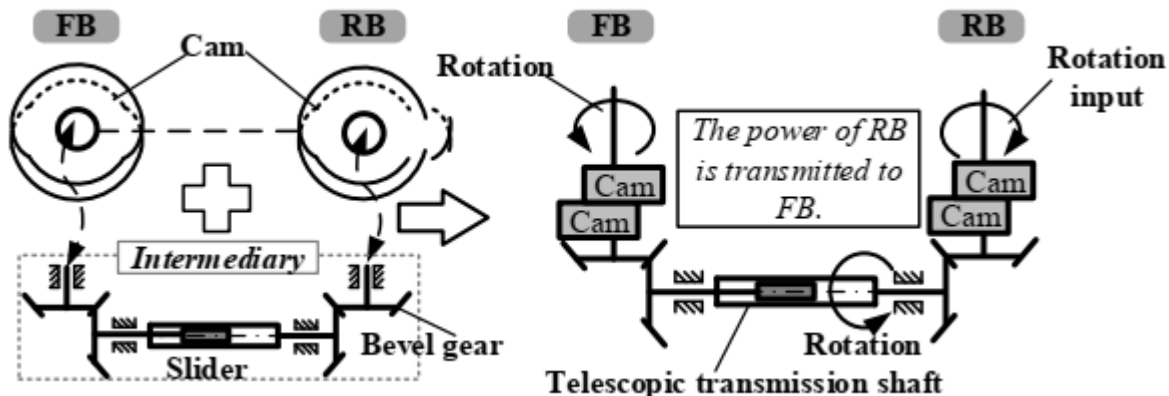
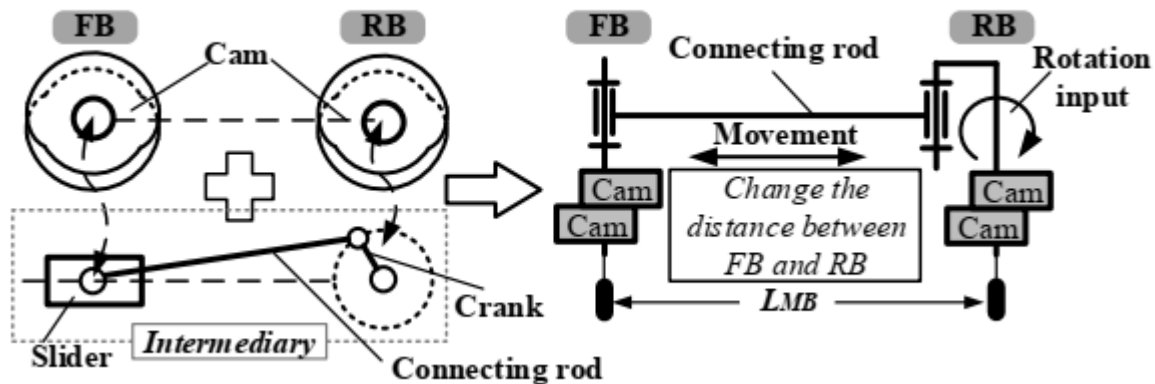
Case study: Conceptual design for an In-Pipe Robot

❑ Resolving Physic conflicts

Nguyên tắc sáng tạo 24 (Trung gian/Môi giới):

Giải pháp 1: Sử dụng cơ cấu tay quay-con trượt để chuyển đổi chuyển động quay thành chuyển động tịnh tiến.

Giải pháp 2: Sử dụng trục truyền động lồng nhau để truyền lực từ RB đến FB.



Case study: Conceptual design for an In-Pipe Robot

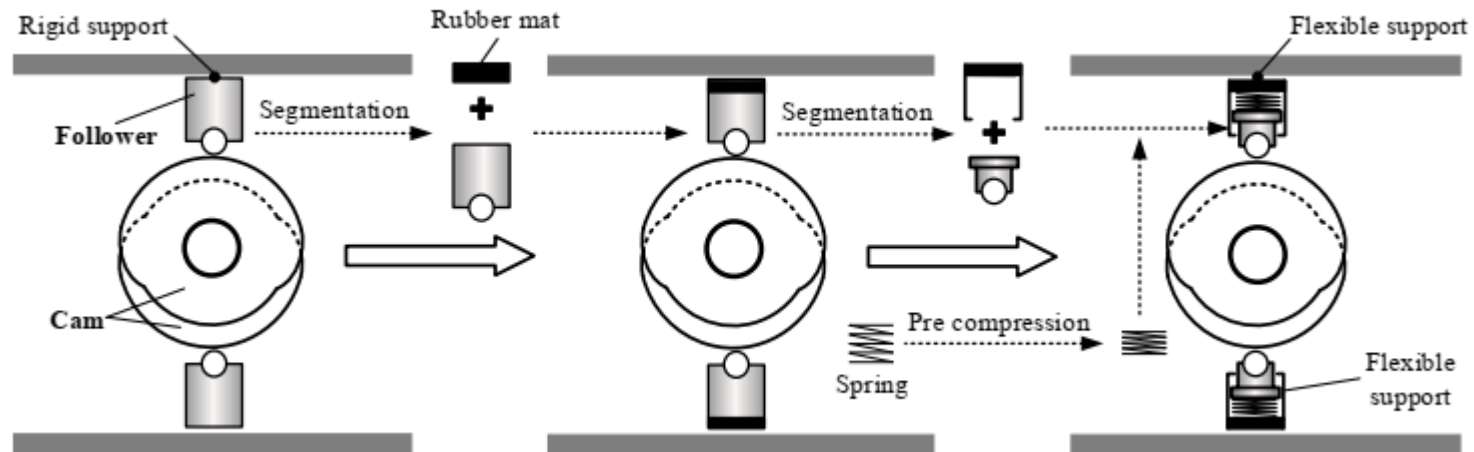
❑ Resolving technical conflicts

Nguyên tắc sáng tạo 1 (Phân đoạn): Chia một đối tượng thành các phần riêng biệt hoặc có thể kết hợp.

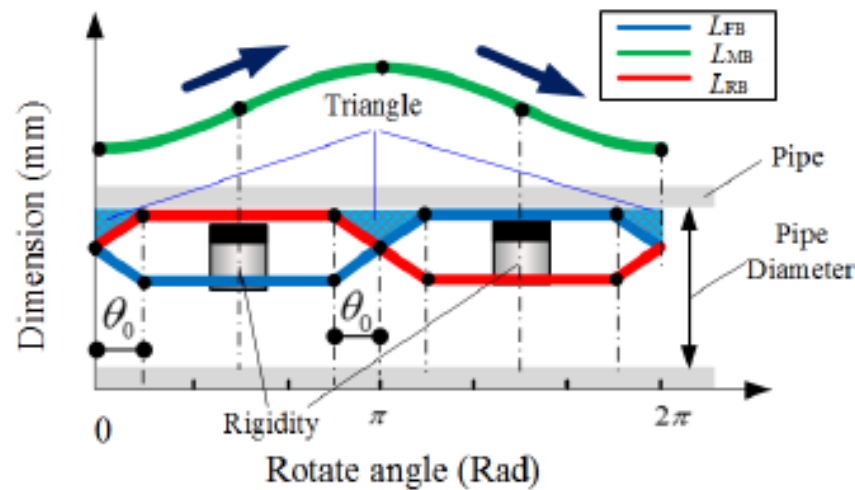
Giải pháp: Thiết kế chân đỡ (bộ theo dõi cam) bao gồm các thành phần cứng và mềm.

Nguyên tắc sáng tạo 11 (Bù trước): Chuẩn bị trước để cải thiện độ tin cậy.

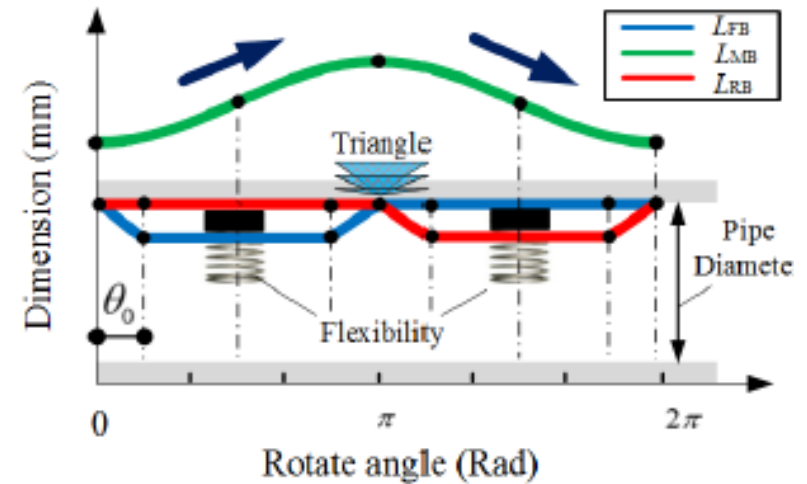
Giải pháp: Thêm lò xo nén trước vào chân đỡ.



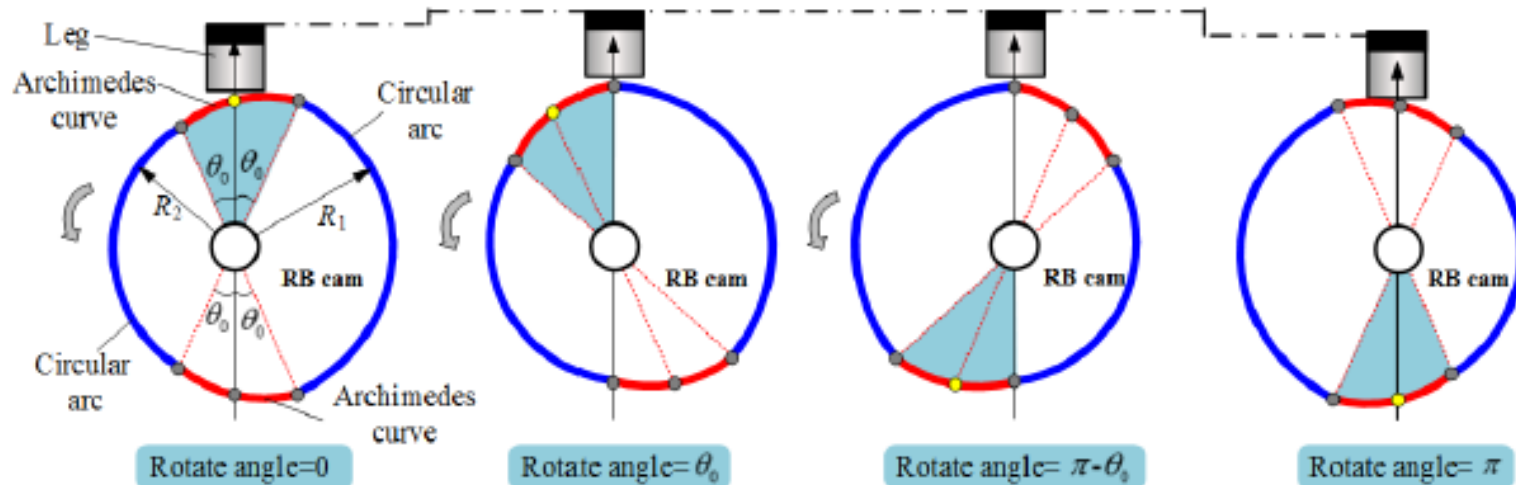
Case study: Conceptual design for an In-Pipe Robot



(a)



(b)



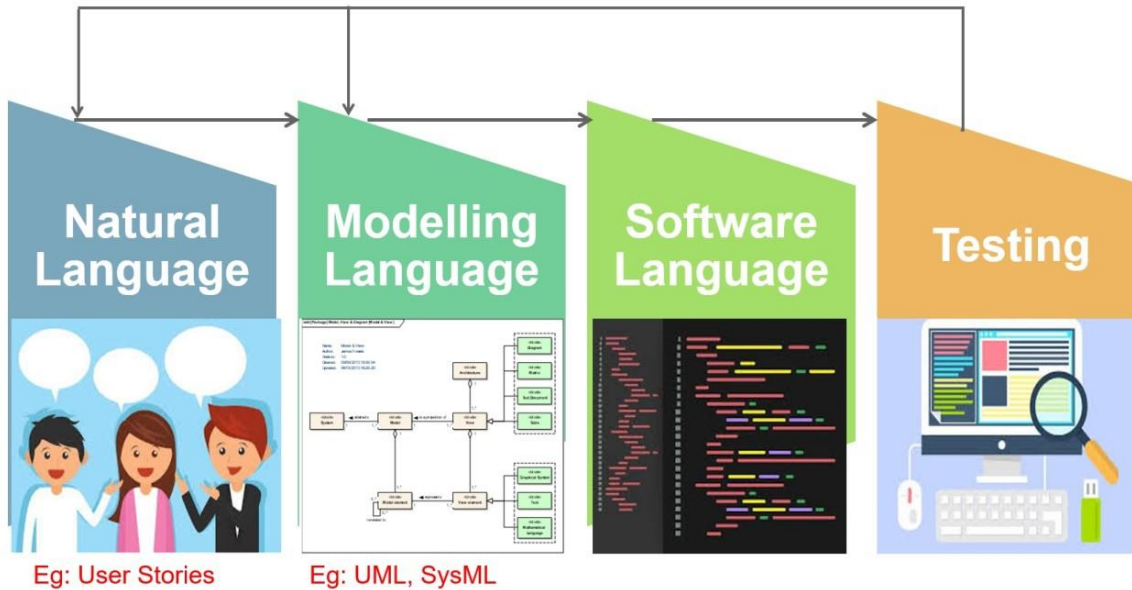
(c)

UML, SysML

UML - Unified Modeling Language

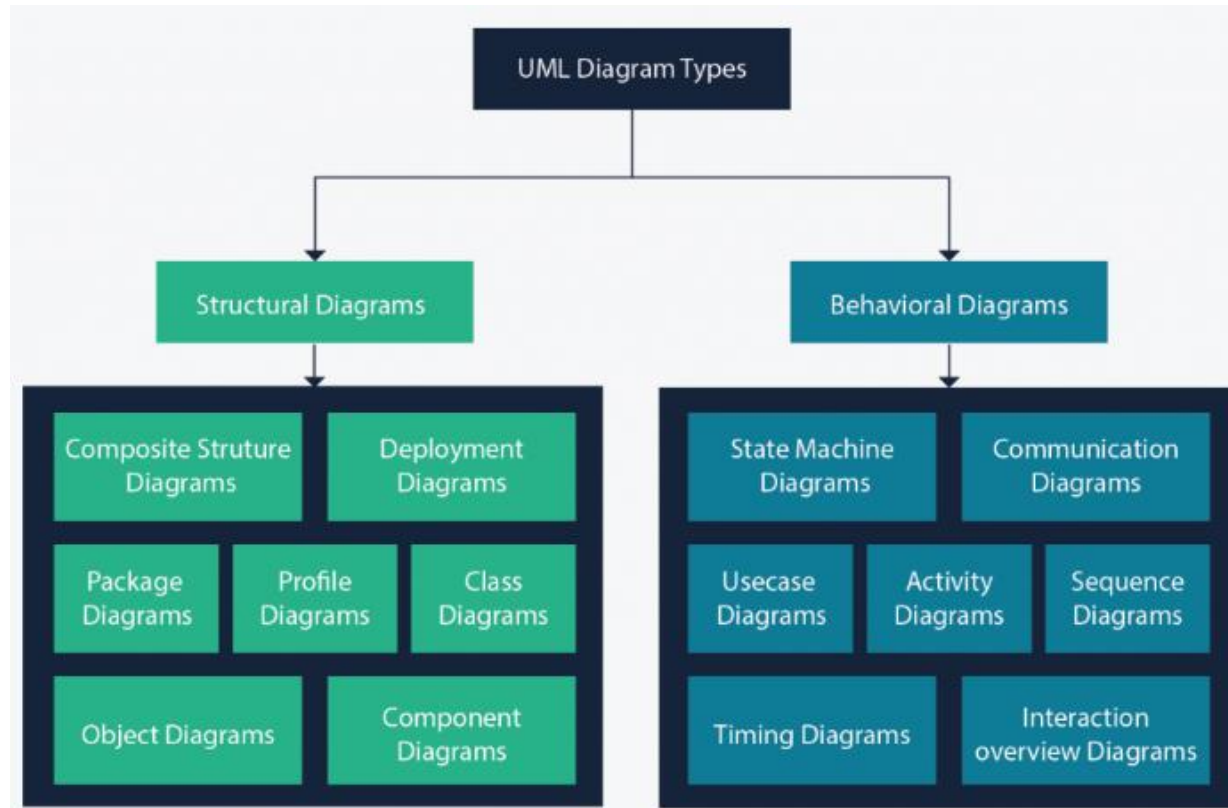
- Unified Modeling Language (UML) consists of several types of diagrams, each serving a specific purpose, to visualize, specify, and document different aspects of **software systems**

MBSE Iterative Process



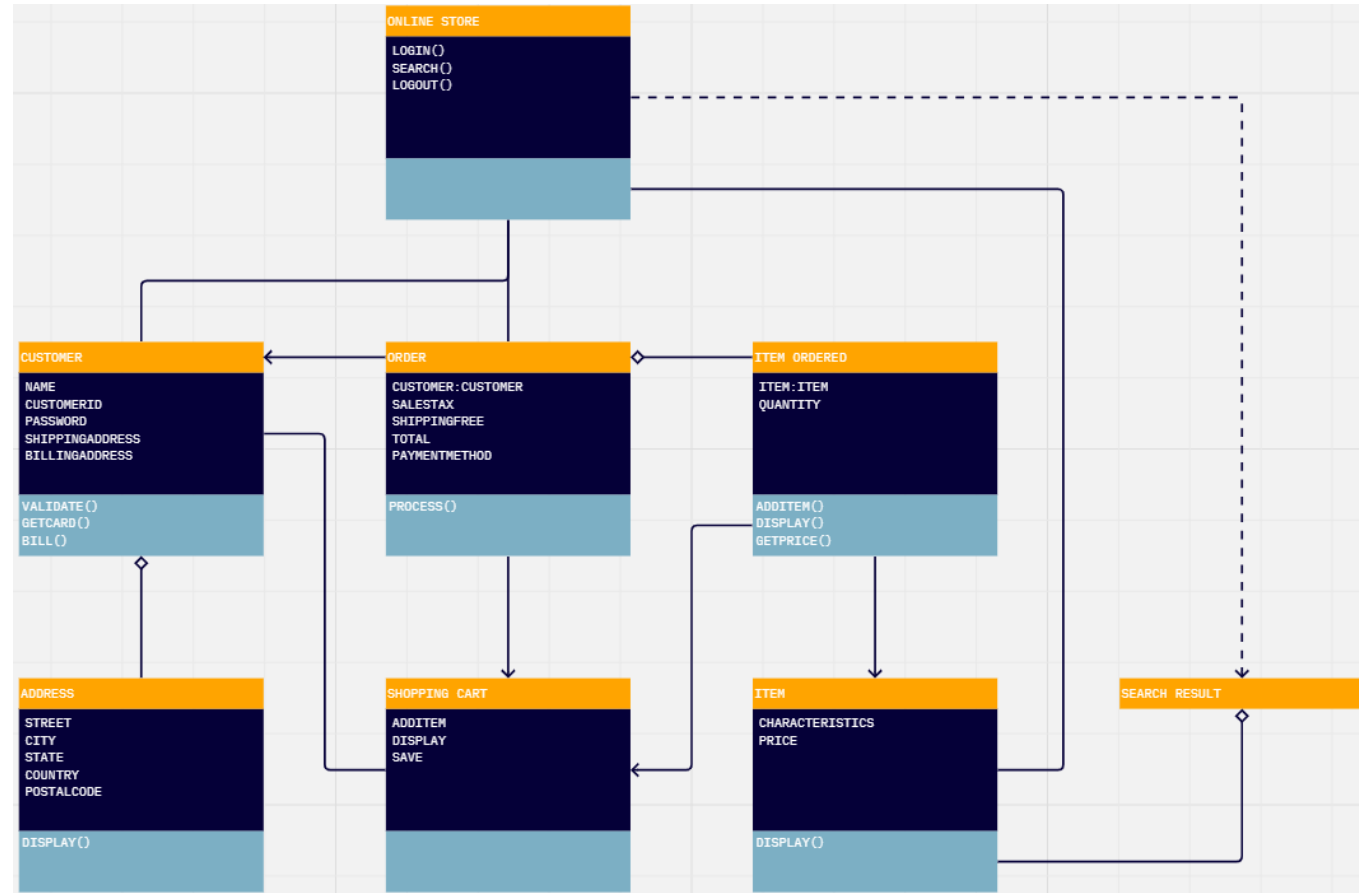
UML - Unified Modeling Language

- **UML diagrams** are divided into two main categories: **Structure Diagrams** and **Behavior Diagrams**, with each category having several diagram types under it.
- **Structure diagrams** Structure diagrams depict the static structure of the system, illustrating the system's classes or objects and their relationships.
- **Behavioral diagrams** Behavior diagrams depict the dynamic aspects of a system, illustrating how the system functions and changes over time.



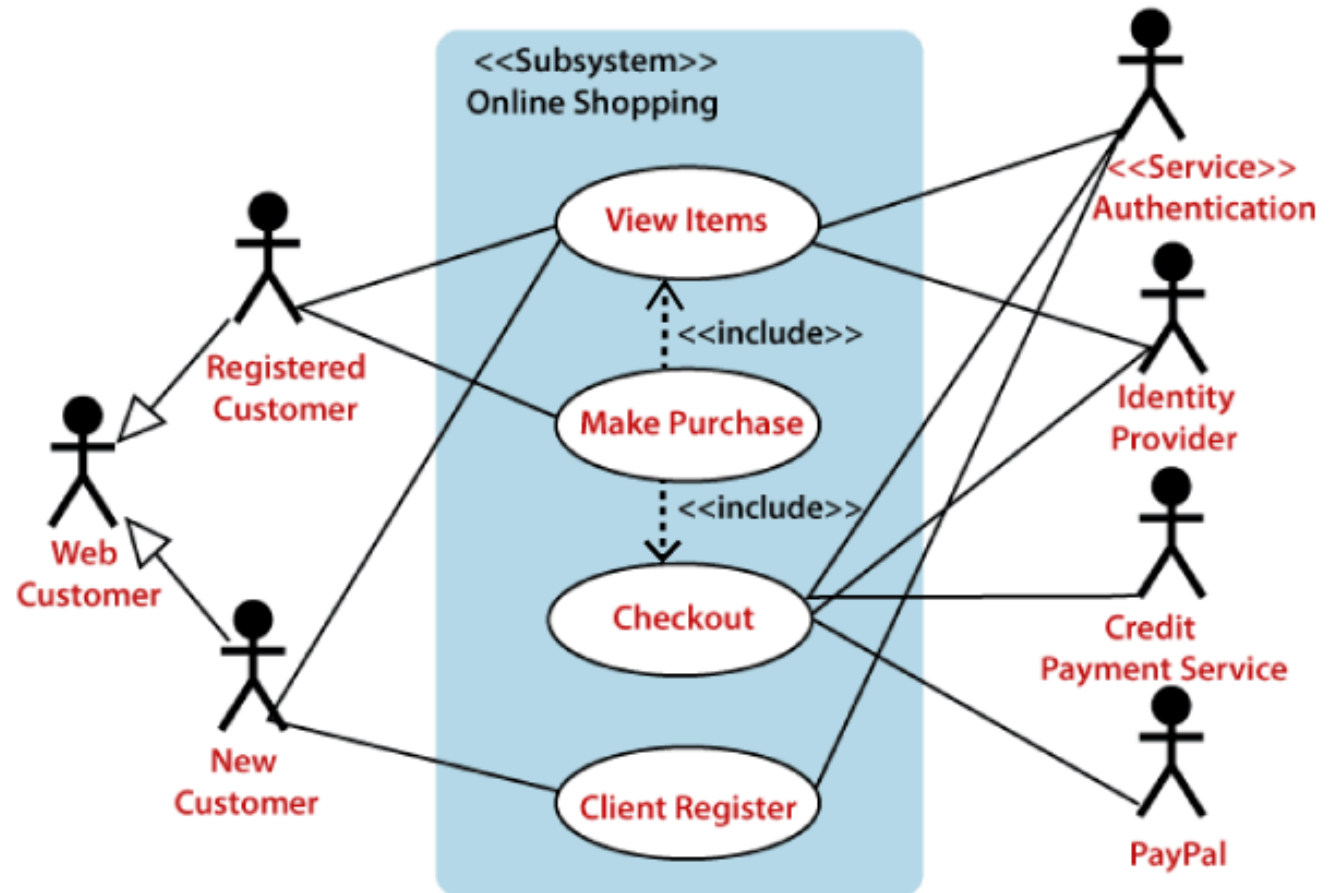
Class Diagram

Represents the static structure of a system, showing classes, attributes, operations, and the relationships between classes.



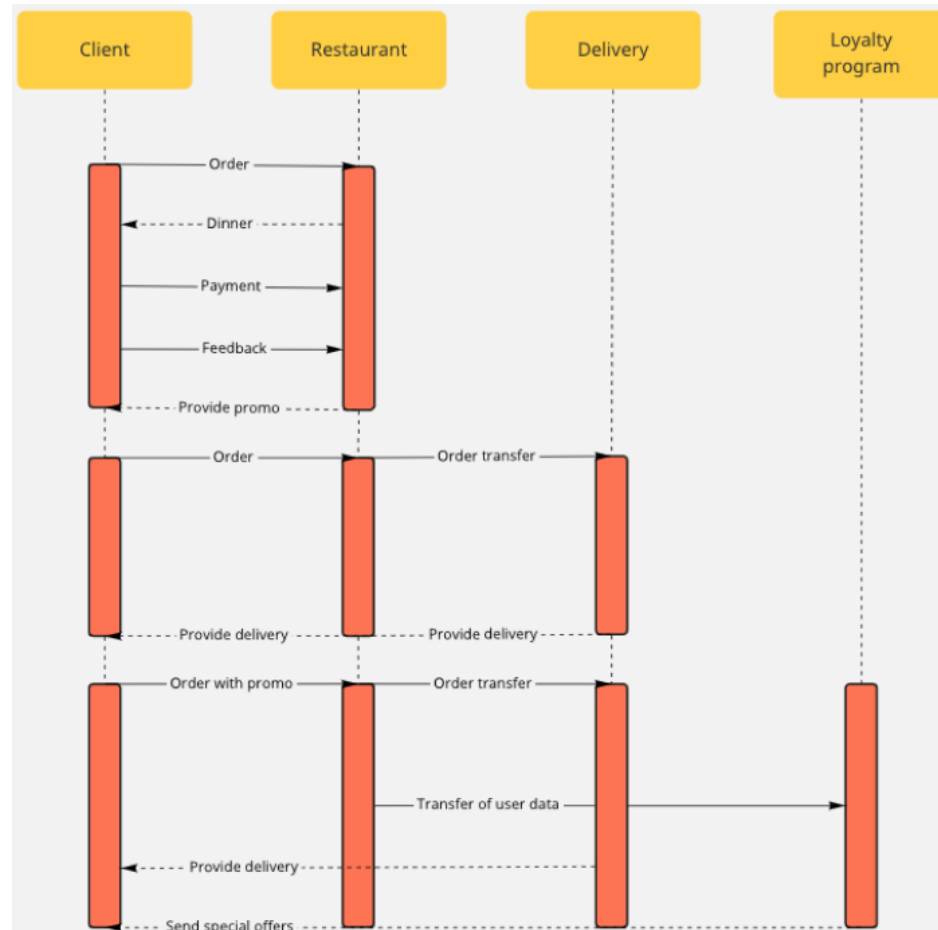
Use Case Diagram

Represents the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases.



Sequence Diagram

Represents the interaction between objects in a specific time sequence.

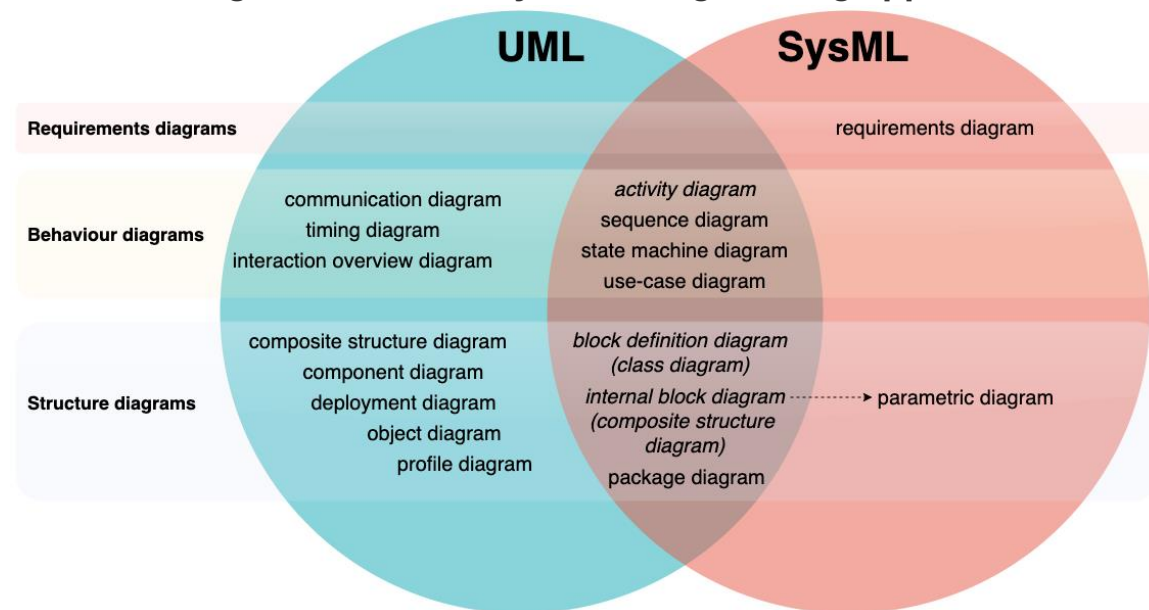


SysML – System Model Language

- SysML is designed to model a **broad range of systems**, which can include hardware, software, data, personnel, and more, making it versatile for various engineering domains.
- **SysML is an extension of UML designed for use in systems engineering applications.**

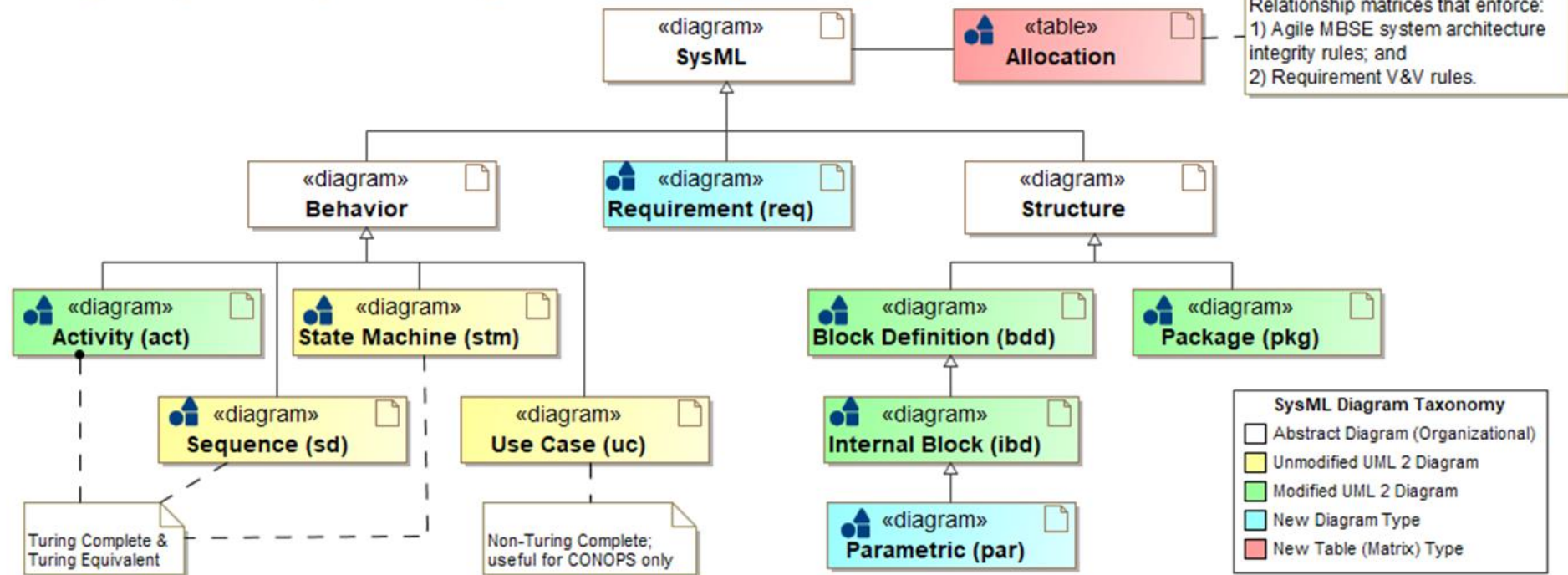
SysML Extensions:

- Definitions for Block Definition and Internal Block Diagrams – BDD & IDB
- Changes in the activity diagram
- Requirements diagram – **New**
- Parametric diagram – **New**
- Allocations (traceability) – **New**



SysML – System Model Language

Agile SysML: SysML Diagram Taxonomy for Agile MBSE™ Methods



SysML Diagrams

